

Translating English to Logic

I did not buy a lottery ticket this week or I bought a lottery ticket and won the million dollar on Friday.

Let p and q be two propositions

p : I bought a lottery ticket this week.

q : I won the million dollar on Friday.

In logic form

$$\neg p \vee (p \wedge q)$$

Conditional Statements

Implication

Definition: Let p and q be propositions. The conditional statement $p \rightarrow q$, is the proposition “If p , then q ”.

The conditional statement $p \rightarrow q$ is false when p is true and q is false and is true otherwise.

where p is called hypothesis, antecedent or premise.

q is called conclusion or consequence

Implication (if - then)

- Binary Operator, Symbol: $\rightarrow$

P	Q	$P \rightarrow Q$
true	true	true
true	false	false
false	true	true
false	false	true

Conditional Statements

Biconditional Statements

Definition: Let p and q be propositions. The *biconditional statement* $p \leftrightarrow q$, is the proposition “ p if and only if q ”.

The *biconditional (bi-implication) statement* $p \leftrightarrow q$ is true when p and q have same truth values and is false otherwise.

Biconditional (if and only if)

- Binary Operator, Symbol: $\leftrightarrow$

P	Q	$P \leftrightarrow Q$
true	true	true
true	false	false
false	true	false
false	false	true

Composite Statements

- Statements and operators can be combined in any way to form new statements.

P	Q	$\neg P$	$\neg Q$	$(\neg P) \vee (\neg Q)$
true	true	false	false	false
true	false	false	true	true
false	true	true	false	true
false	false	true	true	true

Equivalent Statements

P	Q	$\neg(P \wedge Q)$	$(\neg P) \vee (\neg Q)$	$\neg(P \wedge Q) \leftrightarrow (\neg P) \vee (\neg Q)$
true	true	false	false	true
true	false	true	true	true
false	true	true	true	true
false	false	true	true	true

- Two statements are called logically equivalent if and only if (iff) they have identical truth tables
 - The statements $\neg(P \wedge Q)$ and $(\neg P) \vee (\neg Q)$ are **logically equivalent**, because $\neg(P \wedge Q) \leftrightarrow (\neg P) \vee (\neg Q)$ is always true.

Tautologies and Contradictions

- Tautology is a statement that is always true regardless of the truth values of the individual logical variables
- Examples:
- $R \vee (\neg R)$
- $\neg(P \wedge Q) \leftrightarrow (\neg P) \vee (\neg Q)$
- If $S \rightarrow T$ is a tautology, we write $S \Rightarrow T$.
- If $S \leftrightarrow T$ is a tautology, we write $S \Leftrightarrow T$.

Tautologies and Contradictions

- A Contradiction is a statement that is always false regardless of the truth values of the individual logical variables

Examples

- $R \wedge (\neg R)$
- $\neg(\neg(P \wedge Q) \leftrightarrow (\neg P) \vee (\neg Q))$
- The negation of any tautology is a contradiction, and the negation of any contradiction is a tautology.

Exercises

- We already know the following tautology:
- $\neg(P \wedge Q) \Leftrightarrow (\neg P) \vee (\neg Q)$
- Nice home exercise:
- Show that $\neg(P \vee Q) \Leftrightarrow (\neg P) \wedge (\neg Q)$.
- These two tautologies are known as De Morgan's laws.

Logical Equivalence

Definition

Two proposition form are called **logically equivalent** if and only if they have **identical truth values** for each possible substitution of propositions for their proposition variable.

The logical equivalence of proposition forms **P** and **Q** is written

$$P \equiv Q$$

Equivalence of Two Compound Propositions P and Q

1. Construct the truth table for **P**.
2. Construct the truth table for **Q** using the same proposition variables for identical component propositions.
3. Check each combination of truth values of the proposition variables to see whether the truth value of **P** is the same as the truth value of **Q**.

Equivalence Check

- a. If in each row the truth value of P is the **same** as the truth value of Q , then P and Q are **logically equivalent**.
- b. If in some row P has a **different** truth value from Q , then P and Q are **not logically equivalent**.

Example

- Prove that $\neg(\neg p) \equiv p$

Solution

p	$\neg p$	$\neg(\neg p)$
T	F	T
F	T	F

As you can see the corresponding truth values of p and $\neg(\neg p)$ are same, hence **equivalence** is justified.

Example

Show that the proposition forms $\neg(p \wedge q)$ and $\neg p \wedge \neg q$ are **NOT** logically equivalent.

p	q	$\neg p$	$\neg q$	$(p \wedge q)$	$\neg(p \wedge q)$	$\neg p \wedge \neg q$
T	T	F	F	T	F	F
T	F	F	T	F	T	F
F	T	T	F	F	T	F
F	F	T	T	F	T	T

Here the corresponding truth values differ and hence equivalence does not hold

De Morgan's laws

De Morgan's laws state that:

The negation of an **and** proposition is logically equivalent to the **or** proposition in which each component is negated.

The negation of an **or** proposition is logically equivalent to the **and** proposition in which each component is negated.

Symbolically (De Morgan's Laws)

$$1. \neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$2. \neg(p \vee q) \equiv \neg p \wedge \neg q$$

Applying De-Morgan's Law

Question: Negate the following compound Propositions

1. John is six feet tall and he weights at least 200 pounds.
2. The bus was late or Tom's watch was slow.

Solution

- a) John is **not** six feet tall **or** he weighs **less** than 200 pounds.
- b) The bus was **not** late **and** Tom's watch was **not** slow.

Inequalities and De Morgan's Laws

Question Use De Morgan's laws to write the negation of

$$-1 < x \leq 4$$

Solution: The given proposition is equivalent to

$$-1 < x \text{ and } x \leq 4,$$

By De Morgan's laws, the negation is

$$-1 \geq x \text{ or } x > 4.$$

Tautology and Contradiction

Definition A tautology is a proposition form that is always true regardless of the truth values of the individual propositions substituted for its proposition variables. A proposition whose form is a tautology is called a **tautological proposition**.

Definition A contradiction is a proposition form that is always false regardless of the truth values of the individual propositions substituted for its proposition variables. A proposition whose form is a contradiction is called a **contradictory proposition**.

Example

Show that the proposition form $p \vee \neg p$ is a tautology and the proposition form $p \wedge \neg p$ is a contradiction.

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

Exercise: If t is a tautology and c is contradiction, show that $p \vee t \equiv p$ and $p \wedge c \equiv c$?

Laws of Logic

1. Commutative laws

$$p \wedge q \equiv q \wedge p ; \quad p \vee q \equiv q \vee p$$

2. Associative laws

$$p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r ; \quad p \vee (q \vee r) \equiv (p \vee q) \vee r$$

3. Distributive laws

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

Laws of Logic

4. Identity laws

$$p \wedge t \equiv p \quad ; \quad p \vee c \equiv p$$

5. Negation laws

$$p \vee \neg p \equiv t \quad ; \quad p \wedge \neg p \equiv c$$

6. Double negation law

$$\neg(\neg p) \equiv p$$

7. Idempotent laws

$$p \wedge p \equiv p \quad ; \quad p \vee p \equiv p$$

Laws of Logic

8. Universal bound laws

$$p \vee t \equiv t ; p \wedge c \equiv c$$

9. Absorption laws

$$p \wedge (p \vee q) \equiv p ; p \vee (p \wedge q) \equiv p$$

10. Negation of t and c

$$\neg t \equiv c ; \neg c \equiv t$$

Exercise

Using laws of logic, show that

$$\neg(\neg p \wedge q) \wedge (p \vee q) \equiv p.$$

Solution

Take $\neg(\neg p \wedge q) \wedge (p \vee q)$

$$\equiv (\neg(\neg p) \vee \neg q) \wedge (p \vee q), \quad (\text{by De Morgan's laws})$$

$$\equiv (p \vee \neg q) \wedge (p \vee q), \quad (\text{by double negative law})$$

$$\equiv p \vee (\neg q \wedge q), \quad (\text{by distributive law})$$

contd...

$\equiv p \vee (q \wedge \neg q)$, (by the commutative law)

$\equiv p \vee c$, (by the negation law)

$\equiv p$, (by the identity law)

Skill in simplifying proposition forms is useful in constructing logically efficient computer programs and in designing digital circuits.

Another Example

Prove that $\neg[r \vee (q \wedge (\neg r \rightarrow \neg p))] \equiv \neg r \wedge (p \vee \neg q)$

$$\neg[r \vee (q \wedge (\neg r \rightarrow \neg p))]$$

$$\equiv \neg r \wedge \neg(q \wedge (\neg r \rightarrow \neg p)),$$

$$\equiv \neg r \wedge \neg(q \wedge (\neg r \vee \neg p)),$$

$$\equiv \neg r \wedge \neg(q \wedge (r \vee \neg p)),$$

$$\equiv \neg r \wedge (\neg q \vee \neg(r \vee \neg p)),$$

$$\equiv \neg r \wedge (\neg q \vee (\neg r \wedge p)),$$

$$\equiv (\neg r \wedge \neg q) \vee (\neg r \wedge (\neg r \wedge p)),$$

$$\equiv (\neg r \wedge \neg q) \vee ((\neg r \wedge \neg r) \wedge p),$$

$$\equiv (\neg r \wedge \neg q) \vee (\neg r \wedge p),$$

$$\equiv \neg r \wedge (\neg q \vee p),$$

$$\equiv \neg r \wedge (p \vee \neg q),$$

De Morgan's law

Conditional rewritten as disjunction

Double negation law

De Morgan's law

De Morgan's law, double negation

Distributive law

Associative law

Idempotent law

Distributive law

Commutative law